Kyle Werther

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SUMMARY

R&D Engineer with an M.S. in Biomedical Engineering and hands-on medical device experience across design controls, test method development and validation, and GMP/GLP compliant technical documentation under ISO 13485 and 21 CFR Part 820. SOLIDWORKS Certified Professional (CSWP) who designs and prototypes test fixtures, develops ASTM-based test methods, and authors protocols and reports supporting multiple device programs.

EXPERIENCE

Ethicon, Johnson & Johnson MedTech - Raritan, NJ **Apr. 2026 – Present**

R&D Engineer, Performance Evaluation

- Design and prototype 10+ custom SOLIDWORKS fixtures for Instron material testing, iterating designs from concept through validation to characterize device performance and accelerating development timelines
- Perform finite element analysis (FEA) and flow simulations in SOLIDWORKS to predict fixture performance, optimize designs prior to fabrication, and reduce prototyping iterations
- Lead collaboration with external machine shops and suppliers to develop custom testing fixtures, driving design iterations and resolving manufacturing challenges to support device characterization studies
- Engineered a novel mechanical characterization method for nonwoven materials by integrating tensile and compressive testing, transforming an attribute-based assessment into a quantitative variable measurement approach
- Develop and execute ASTM-based mechanical test methods on Instron / Bluehill Universal supporting design changes, test method validation, and new product development across 5 biosurgery device programs
- Author test method validation protocols, reports, and engineering memos, applying Minitab statistical analysis (Gage R&R, ANOVA, regression) to support design decisions across multiple design change projects
- Execute in vitro blood interaction studies using Nikon microscopy to evaluate product performance and generate data supporting device development and verification activities

Doctor Androulakis Research Group - Rutgers University **May 2024 – Apr. 2026**

Graduate Researcher

- Expanded an in-silico circadian rhythm model to incorporate the inflammatory response system, calibrating 70+ parameters across 22 nonlinear ODEs in MATLAB against in-vitro experimental datasets
- Communicated computational findings to multidisciplinary audiences through authoring and presenting thesis chapters
- Identified various parameters sensitivities driving model behavior through simulating various temperature states

Cresilon - Brooklyn, NY **Jan. 2023 – May 2023**

Quality Assurance Intern

- Supported 510(k) clearance of TraumaGel, a Class II hemostatic device now commercially available, by drafting 3 design verification protocols and 2 final reports under 21 CFR 820 design controls
- Established a seal integrity SOP by conducting a root cause analysis to further develop and refine test methodology
- Audited batch release documentation across multiple critical procedures to verify safety and compliance standards

EDUCATION

Rutgers University - New Brunswick, School of Engineering

Master of Science in Biomedical Engineering **May 2026**

Bachelor of Science in Biomedical Engineering **May 2024**

SKILLS

Technical: Bluehill Software, Instron, Design Controls, Design of Experiments (DOE), ISO 13485, 21 CFR Part 820, GLP, GMP, MATLAB, Microsoft Suite, Minitab, Nikon Microscopy, Python, Root Cause Analysis (RCA), SOLIDWORKS (3D CAD), Technical Writing, Test Method Development/Validation

Certifications: AIGPE Certified Minitab Expert, SOLIDWORKS Certified Professional (CSWP)